

Clinical Evidence Review

Case BV-225

Generated 2026-10-07 16:09:35

Study Medical image · 1024×1024

Current presentation

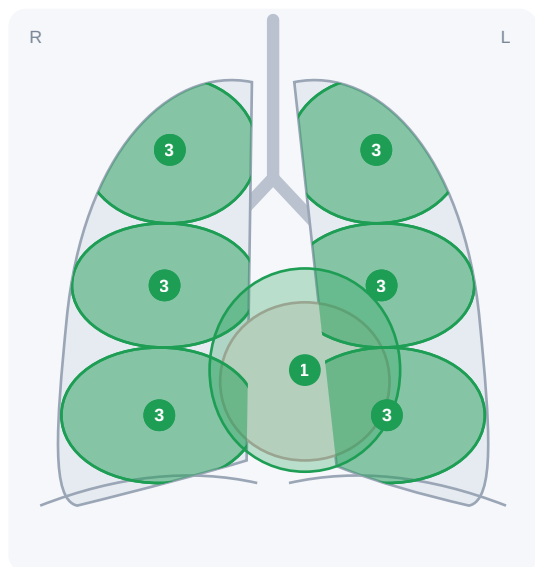
Tired all the time for 3 weeks, breathless on climbing one flight of stairs, both ankles swollen by evening.

Present: Fatigue (3 weeks), Shortness of breath, Ankle / leg swelling **Denied:** —

Relevant patient history

2026-09-03	Cardiomyopathy	Echo report · history.pdf · page 1
2026-09-03	Diuretic therapy	Echo report · history.pdf · page 1
2026-09-01	Valvular heart disease	Echo report · history.pdf · page 2
2010	Hypertension	Echo report · history.pdf · page 1

Where the findings are



1 Cardiomegaly — supported

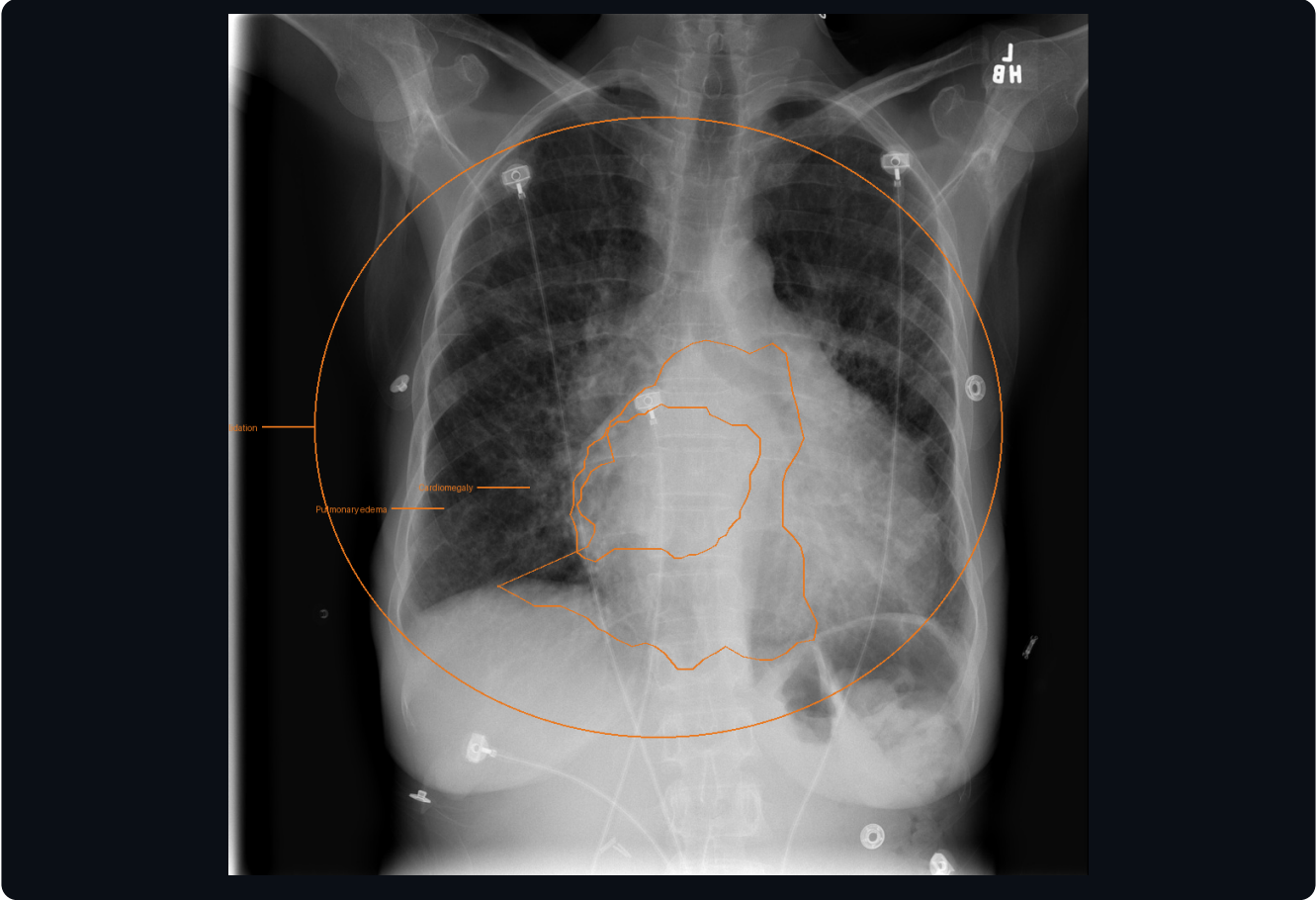
cardiac silhouette · model outline

2 Pulmonary edema — supported

right upper zone, right mid zone, right lower zone, left upper zone, left mid zone, left lower zone · model outline

3 Consolidation — supported

right upper zone, right mid zone, right lower zone, left upper zone, left mid zone, left lower zone · approximate zone



3 candidate finding(s): 3 supported. Overall evidence quality: **HIGH**.

Finding 01

Cardiomegaly

Supported

Consider cardiomegaly (Heart). Imaging and the supplied context agree.

IMAGE EVIDENCE	RELEVANT HISTORY	CURRENT PRESENTATION
• The heart appears enlarged, with a prominent cardiac silhouette. • Primary image signal: strong • Independent verifier: strong • “chf with interstitial and question alveolar edema, slightly improved” • “chf, with vascular blurring and scattered foci of alveolar edema” • Localized to Heart · segmented	• Cardiomyopathy · 2026-09-03 Echo report · history.pdf · page 1 • Valvular heart disease · 2026-09-01 Echo report · history.pdf · page 2 • Hypertension · 2010 Echo report · history.pdf · page 1	• Fatigue · 3 weeks • Shortness of breath • Ankle / leg swelling

CONFLICTING / NEGATIVE EVIDENCE

None identified

EVIDENCE STRENGTH

HIGH

Image confidence 96 % · CLEAR 99 %,
CheXzero 94 %
calibrated on 202 labelled films (CheXpert)

Finding 02 Pulmonary edema

Supported

Consider pulmonary edema (Lung). Imaging and the supplied context agree.

IMAGE EVIDENCE

- There is increased opacity in the lung fields, suggestive of pulmonary edema.
- Primary image signal: strong
- Independent verifier: strong
- “chf with interstitial and question alveolar edema, slightly improved”
- “chf, with vascular blurring and scattered foci of alveolar edema”
- Localized to Lung · segmented

RELEVANT HISTORY

- Cardiomyopathy · 2026-09-03
Echo report · history.pdf · page 1
- Diuretic therapy · 2026-09-03
Echo report · history.pdf · page 1
- Valvular heart disease · 2026-09-01
Echo report · history.pdf · page 2

CURRENT PRESENTATION

- Shortness of breath
- Ankle / leg swelling

CONFLICTING / NEGATIVE EVIDENCE

- Film also resembles “no substantial change or slight worsening of extensive interstitial pulmonary edema with some alveolar components”

EVIDENCE STRENGTH

HIGH

Image confidence 82 % · CLEAR 92 %,
CheXzero 73 %
calibrated on 202 labelled films (CheXpert)

Finding 03 Consolidation

Supported

Consider consolidation (Lung). Imaging and the supplied context agree. The visual reasoning model did not describe it — check the film.

IMAGE EVIDENCE

- Visual reasoning model did not identify this finding
- Primary image signal: strong
- Independent verifier: moderate
- “pneumonia or edema superimposed upon chronic lung disease”
- “suspicious for pneumonia with superimposed pulmonary edema”

RELEVANT HISTORY

None identified

CURRENT PRESENTATION

- Shortness of breath

CONFLICTING / NEGATIVE EVIDENCE

None identified

EVIDENCE STRENGTH

HIGH

Image confidence 32 % · CLEAR 48 %,
CheXzero 17 %
calibrated on 202 labelled films (CheXpert)

Checked and rejected

- Right upper lobe opacity MedGemma 1.5 — CLEAR, reading the same phrase against the film, disagrees (0.79 < 0.85)
- Left lower lobe opacity MedGemma 1.5 — CLEAR, reading the same phrase against the film, disagrees (0.63 < 0.85)
- Right-sided central venous catheter MedGemma 1.5 — CLEAR, reading the same phrase against the film, disagrees (0.79 < 0.85)
- Left-sided central venous catheter MedGemma 1.5 — CLEAR, reading the same phrase against the film, disagrees (0.77 < 0.85)
- Elevated right hemidiaphragm MedGemma 1.5 — CLEAR, reading the same phrase against the film, disagrees (0.59 < 0.85)

- **Pleural effusion** CLEAR + CheXzero — prompt score was high, but no phrase among the film's top 300 of 368,294 report phrases names it (best #1022)
- **Atelectasis** CLEAR + CheXzero — prompt score was high, but no phrase among the film's top 300 of 368,294 report phrases names it (best #775)
- **Pulmonary fibrosis** CLEAR concepts — the film's closest report phrases suggest it, but MedGemma did not confirm it on the image
- **Lines & devices** CLEAR + CheXzero — prompt score was high, but no phrase among the film's top 300 of 368,294 report phrases names it (best #545)
- **Widened mediastinum** label hierarchy — already explained by cardiomegaly

Limitations

Findings are candidate observations generated by automated models and reconciled against the supplied documents. Evidence strength is a rule-based summary of agreement between sources. Image confidence is each image reader's probability calibrated (Platt scaling) on labelled films, and reflects the image only; the prevalence in clinical use may differ from the calibration set. Localization is approximate. Models used: CLEAR (primary), CheXzero (verifier), MedGemma 1.5 4B (localization and visual description).

Review statement

This report is generated as clinical decision support from the supplied imaging and contextual evidence. It is not a definitive diagnosis and requires review by a qualified clinician.